

NORTHERN COLORADO, CO, USA

WMO: 724769

Lat: 40.450N

Lon: 105.017W

Elev: 5015

StdP: 12.22

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 94035

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	0.0	5.5	-7.9	4.3	2.2	-1.6	6.1	12.6	30.9	34.6	26.8	37.7	4.6	20	0.563

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	28.4	94.8	61.1	91.3	61.2	89.5	61.0	65.9	83.1	64.6	82.2	63.4	81.5	9.1	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.3	97.6	70.3	59.2	90.5	69.8	57.1	84.0	69.1	33.5	83.6	32.4	82.3	31.4	80.8	71.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 25.8	21.3	18.0	DB	-9.0	99.6	5.5	2.4	-13.0	101.4	-16.2	102.8	-19.3	104.1	-23.3	105.9	(4)
(5)			WB	-9.7	68.1	5.4	1.3	-13.5	69.1	-16.7	69.9	-19.7	70.6	-23.6	71.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	50.2	29.8	31.8	41.2	47.6	56.8	67.4	73.7	71.3	63.4	50.0	38.7	29.3		
	DBStd	17.84	10.52	10.83	9.77	8.72	8.04	6.62	4.94	4.64	7.71	9.15	9.78	10.31		
	HDD50	2726	627	512	296	144	33	1	0	0	8	111	352	642		
	HDD65	6096	1091	930	738	522	271	50	5	9	115	468	790	1107		
	CDD50	2787	0	2	23	72	242	523	734	659	410	110	12	0		
	CDD65	683	0	0	0	0	15	122	274	203	67	2	0	0		
	CDH74	8963	0	0	10	53	403	1753	3236	2340	1049	117	2	0		
	CDH80	3936	0	0	0	3	113	764	1618	1026	394	18	0	0		
(14) Wind	WSAvg	7.1	6.3	7.2	8.0	9.0	7.7	7.2	6.8	6.6	6.5	6.7	6.4	6.4		
(15) Precipitation	PrecAvg	16.3	0.4	0.6	1.3	2.2	2.8	1.7	1.7	1.5	1.7	1.2	0.7	0.6		
	PrecMax	22.0	1.5	1.5	3.9	6.6	7.8	3.8	3.9	3.7	8.2	2.6	1.7	2.3		
	PrecMin	8.6	0.0	0.2	0.0	0.2	0.9	0.0	0.4	0.2	0.0	0.2	0.2	0.2		
	PrecStd	3.4	0.4	0.4	1.0	1.3	1.5	1.0	0.9	0.9	1.5	0.7	0.4	0.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	62.9	65.8	74.9	79.4	88.7	97.5	99.4	95.4	93.0	82.1	72.0	60.7		
		MCWB	42.8	44.8	48.8	51.1	57.6	60.9	62.7	61.1	59.8	54.6	48.5	42.3		
	2%	DB	55.3	59.4	69.8	74.8	82.3	92.6	95.8	91.8	88.5	77.3	65.6	54.7		
		MCWB	39.8	41.9	46.9	50.1	55.9	59.9	61.6	61.3	58.8	52.7	46.3	39.8		
	5%	DB	51.6	54.5	64.1	71.4	78.9	88.4	92.7	89.9	84.3	72.3	61.2	49.9		
		MCWB	38.1	39.5	44.7	48.9	55.1	59.5	61.8	61.3	57.9	51.1	44.2	37.2		
	10%	DB	45.9	48.4	59.2	65.5	73.2	83.9	90.0	86.3	81.0	66.3	55.1	45.4		
		MCWB	35.1	36.5	42.5	47.1	53.6	59.1	62.0	61.1	57.2	48.9	41.1	34.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	44.0	45.5	50.2	53.9	61.9	66.4	68.2	66.7	64.6	56.5	49.5	43.2		
		MCDB	60.2	63.5	72.0	73.5	79.7	82.4	85.5	84.0	82.4	76.4	68.6	59.2		
	2%	WB	40.6	42.3	47.6	51.4	59.0	64.1	66.4	65.1	61.8	54.1	46.9	40.2		
		MCDB	54.9	58.5	66.9	71.0	75.3	82.0	83.9	82.3	80.3	72.6	64.2	53.8		
	5%	WB	37.9	39.7	45.3	49.8	56.9	62.2	65.2	63.9	60.0	52.0	44.5	37.4		
		MCDB	50.0	53.5	63.3	68.4	72.6	81.1	82.7	81.7	77.4	69.1	60.2	48.9		
	10%	WB	35.4	36.9	43.0	47.8	55.1	60.5	64.0	62.6	58.5	50.1	41.8	35.0		
		MCDB	45.5	48.3	58.3	64.0	71.1	79.1	81.9	80.3	75.4	65.2	54.8	44.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	23.8	23.8	26.7	25.5	25.6	28.4	28.4	28.0	27.8	25.6	24.3	22.8		
		MCDBR	31.4	32.1	34.8	35.6	34.3	35.9	34.4	33.6	34.9	34.9	32.7	29.9		
		MCWBR	19.0	18.3	17.8	16.9	14.1	12.7	11.1	11.0	12.8	15.6	17.7	18.3		
	5% WB	MCDBR	29.8	30.6	33.5	32.4	29.1	30.3	28.1	28.3	29.4	30.6	31.8	28.7		
		MCWBR	18.6	17.9	17.3	16.0	13.3	12.4	10.4	10.6	11.8	14.3	17.5	18.0		
(40) Clear-Sky Solar Irradiance	taub		0.244	0.253	0.277	0.304	0.327	0.334	0.364	0.357	0.315	0.277	0.258	0.248		
	taud		2.596	2.579	2.522	2.472	2.464	2.461	2.420	2.429	2.532	2.615	2.582	2.579		
	Ebn at Noon		302	313	313	308	301	298	287	287	295	299	292	291		
	Edh at Noon		21	25	30	34	35	35	36	35	29	24	21	20		
(44) All-Sky Solar Radiation	RadAvg		785	1094	1487	1738	1923	2257	2133	1876	1619	1165	829	668		
	RadStd		38	68	115	119	160	121	89	70	91	79	51	46		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.76	N/A	N/A	N/A	+0.99	N/A	N/A	N/A	N/A	N/A
(47) Regional (29 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page